

Order of Operations: Lecture Hall Fire Scenario

An AccessControlSystem is created to act as the receiver, along with a CampusComponent representing the lecture hall area. A SecureAreaCommand is then created, linking the access control system to that area.

A CommunicationService is created for the lecture staff, acting as a FacilityStaff. Its state is changed once, moving it into Detection. At this point the staff does not yet have a command assigned to it.

A second CommunicationService is created for hall security, acting as a CampusSecurity. Its state is also changed once, placing it into Detection.

An IssueAlertCommand is created, linking the security receiver to the issue message "One of the lectureHalls is on fire".

An EmergencyCoordinator is created to mediate between staff and security. The staff's mediator is set to this coordinator, and security is registered as a component the staff can notify. The staff's mediator then notifies security directly, alerting it to the fire.

A CommunicationService is created for the medical responders. Its state is changed once, moving it into Detection, and an IssueAlertCommand is created linking the medical responders to the same issue message.

A second EmergencyCoordinator is created for security's side. Staff, the building (access control), and the medical responders are all registered as components security can notify.

Security's state is changed from Detection to Response, and security's mediator notifies all three registered components of this change.

The medical responders' state changes, followed by the staff's state changing again, each continuing their own progression through the sequence.

The operator console executes three commands in turn: the security alert, the lecture hall security management command, and the medical alert.

Finally, all heap-allocated objects created directly by the invoker and the mediators are deleted: the operator console and both mediators.